符号	含义	对应中文解释
RP	Rephrase	重新表述（语病或表达不当）
RM	Remove	删除
Δ	Define / Explain	定义或详细解释
Ref(s)	Reference(s) needed	需要补充参考文献
M.I.	Missing Information	信息缺失
?	Question	疑问/不明确

标题

note 和作者需要 RM（删除）

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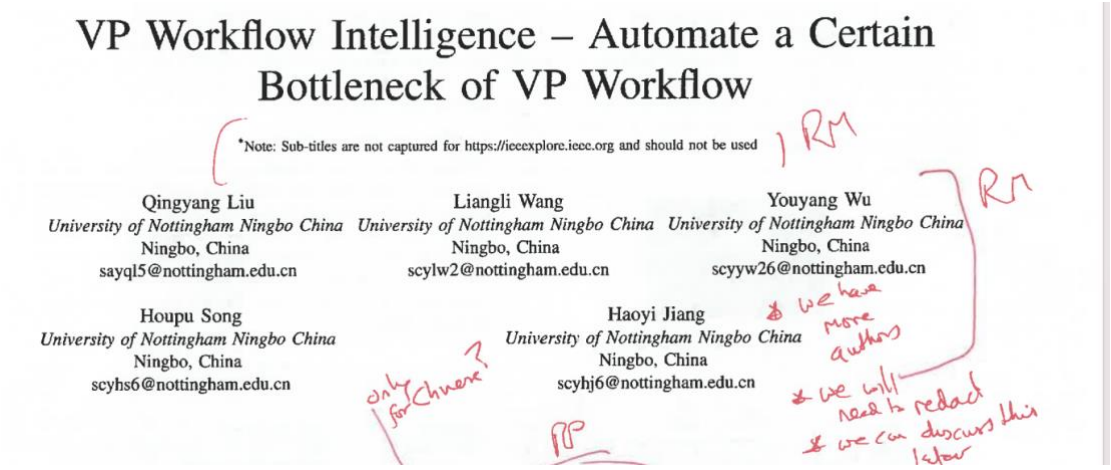

VP Workflow Intelligence – Automate a Certain Bottleneck of VP Workflow

*Note: Sub-titles are not captured for https://ieeexplore.ieee.org and should not be used

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Abstract—Virtual Production (VP) transforms filmmaking by replacing physical set construction with real-time digital environments, while Artificial Intelligence (AI) addresses bottlenecks in 3D asset creation and technical optimization—critical for Chinese short films pursuing international appeal. This paper presents a scalable AI-VP integration framework developed for a university-led short film production, centered on In-Camera Visual Effects (ICVFX) and generative AI. We detail requirements engineering, agile system design, and prototype development that meet industry ICVFX standards. The framework reduces AI-assisted asset creation time by over 60% and resolves cross-disciplinary collaboration gaps between software engineering and filmmaking teams. Empirical validation confirms the framework's feasibility for small/medium film teams, enabling Chinese short films to achieve international production value with limited resources.

Index Terms—Virtual Production; ICVFX; Generative AI; 3D Asset Creation; Chinese Short Films; Workflow Integration

I. Introduction

introduction

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* Do not use GuAl to create!（不要直接用 AI 生成内容/格式）以及 #Why so much bold fout?（为什么用这么多加粗字体？）

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I. Introduction

Smart systems have revolutionized cross-disciplinary industrial applications by integrating context awareness, adaptive control, and real-time decision-making into software frameworks—an evolution that is reshaping the filmmaking industry into **smart filmmaking**. Unlike traditional filmmaking, smart filmmaking leverages cyber-physical systems (CPS) to unify physical filming equipment and digital virtual environments, demanding software solutions that can dynamically adapt to creative requirements, environmental context (e.g., camera position, lighting conditions), and technical constraints. For Chinese short film crews pursuing international appeal, the adoption of smart filmmaking is hindered by two core challenges: the lack of adaptive software frameworks for virtual production (VP) and the technical complexity of aligning AI-driven asset creation with real-time ICVFX workflows.

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Refs

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中间一段的加粗字体后面的一句话 特别指出了“首个为。。。电影创作的框架”这句话是真的么 应该也是因为缺乏文献的支持

Our stakeholders—a professional film crew with award-winning traditional short film productions (e.g., *Half the World s Sleeping*)—sought to transition to smart filmmaking by ntegrating VP and AI but lacked a tailored software engi-neering framework to support this shift. This paper reports n a collaborative project between a software engineering eam and the film crew to design, implement, and validate a **ontext-aware adaptive software framework for AI-driven VP**—the first such framework tailored for smart filmmaking n Chinese short film production. The team was embedded nto the production workflow as smart system developers, vith two core objectives: (1) to deliver a context-adaptive irtual production software stack that meets the real-time and

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Background research

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A. VP and ICVFX Core Workflow

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VP shifts filmmaking from fix-it-in-post to real-time physical-digital convergence, with ICVFX as its core [1]. ICVFX relies on three components: (1) real-time camera

详细解释 natural parallax effect

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这里关于 fig1 提出 explain more 和 ref 对 fig1 进行更多的解释

tracking (OptiTrack/Vicon) for lens/position data, (2) UE-synchronized rendering for natural parallax effects (surpassing green screens) [2], and (3) professional hardware integration (ARRI Alexa 35 camera + LED walls) for color/lighting consistency. ICVFX requires pre-production-focused Virtual Art Department (VAD) to deliver high-fidelity 3D assets running at ≥24fps for LED walls. Fig. 1 is the In-Camera VFX Editor software.

Fig. 1. In-Camera VFX Editor software.

B

重新定义/补偿文献 关于 VP 3d asset bottleneck

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to creative storytelling—critically for international appeal.

C. Industry Case Studies

1. The Mandalorian: Fig.2 shows the virtual production system for The Mandalorian. UE-powered StageCraft LED walls use lighting from the source (digital environment light emitted onto actors), eliminating green spill and enabling natural physical-digital integration [6]. This emphasizes high-contrast lighting elements in virtual set design.

2. Everything Everywhere All At Once: A 5-person VFX team delivered 500 shots via AI-assisted repetitive task automation (rotoscoping/background removal) [7], validating AI's ability to scale small-team output to industry levels—core to this project's design.

Fig. 2. Virtual production System.

REQUIREMENTS ENGINEERING 需求工程

A three-dimensional feasibility study 与 UE5 需要 △ 定义/详细解释

III. REQUIREMENTS ENGINEERING

A. Feasibility Study

A three-dimensional feasibility study confirmed project viability: (1) Technical: Core tasks (UE5 asset creation, ICVFX optimization) use reusable prior project assets; Tripo AI generates compliant base assets. (2) Device: University VP studio

关于 VP 的电脑配置有一个 ？ 整段需要 Refs 需要补充文献

修改一下那两个” 格式错误 B 段需要 RP 和 Refs

provides ARRI Alexa 35, LED walls, and high-performance workstations (I7-14700KF, RTX4060, 32GB DDR5). (3) Legal: Paid AI tool users hold full IP rights to assets; open-source Blender complies with GNU GPL [8]; reverse-search mitigates infringement risks.

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B. Elicitation and Specification

Requirements were elicited via 2 formal stakeholder meetings and weekly film crew production check-ins, translating creative/abstract language (e.g., "soft lighting") into technical specs (e.g., "10% translucent glass for light diffusion"). A signed Software Requirements Specification (SRS) defined functional (virtual set/asset design) and non-functional ICVFX standards (Table 1), aligned with the film crew's January 2026 filming timeline.

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table I 被标记的 应该改成 table 1 （我也不清楚他圈出来是这个意思么）

Outward-facing normals 被标记了 ？

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TABLE I
KEY NON-FUNCTIONAL ICVFX REQUIREMENTS

Category	Core Requirement
Performance	8,000–12,000 draw calls/frame (on-set); ≥50% triangle reduction per LOD
Usability	Centimeter units; English alphanumeric naming; outward-facing normals
Compatibility	FBX format (Z-up); UE5.5.x integration; UCX/UBX/USP collision naming
Maintainability	PBR texture channels only; clean mesh topology

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C. Verification

The SRS was formally verified and signed by the film crew's stakeholder representative and Technical Equipment Team (TET) on November 28, 2025, confirming deliverable deadlines (e.g., window asset prototype by November 24) and alignment with VP equipment testing schedules.

is this important?

System design and prototype development

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关于 virtual set prototype 这一块

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IV. SYSTEM DESIGN AND PROTOTYPE DEVELOPMENT

A. Agile Design and Hardware/Software Stack

Agile development was adopted for dynamic filmmaking requirements: weekly sprints aligned with crew meetings, Monday.com Kanban/Gantt for real-time task tracking, and a continuous feedback loop to translate creative input into high-priority technical tasks.

A unified hardware/software stack (Table 2) was selected for VP industry adoption and AI integration, with Windows 11 (UE/TET compatibility) and high-performance workstations as the base. UE5.5.4 is the core, leveraging Nanite (real-time high-polygon rendering) and Lumen (dynamic global illumination) for ICVFX photorealism [9].

B. Prototype Development and Validation

1. Virtual Set Prototype: A minimalist modern quarantine room (the short film's primary set) was developed in UE5, with light-colored walls, tiled flooring, a central bed, and curtain-divided space. Key features: natural window lighting (core light source) and Blueprint-controlled curtain fluttering (triggered by door opening) for interactive realism.

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TABLE II
CORE SOFTWARE STACK FOR AI-VP FRAMEWORK

Software	Function	Key Advantage
UE5.5.4	ICVFX rendering/virtual set construction	Industry VP standard; Blueprint visual scripting
Blender	High-complexity 3D asset modeling	Free/open-source; comprehensive toolset
Tripo AI	Generative 2D-to-3D asset creation	60%+ reduction in asset creation time
Perforce	Version control	VP-optimized for large binary 3D assets

T17?

2. Core Asset Prototype: Window: The window (critical for ICVFX lighting from the source) was developed in Blender/Tripo AI with two SRS-compliant modifications: (1) non-reflective white/gray silver-bar frame (no harsh filming light reflections); (2) slightly stained 10% translucent glass (soft light diffusion). The prototype met all ICVFX standards: ¡1,000 draw calls/frame, 65% LOD triangle reduction, and UE5 FBX import compatibility.

3. Validation: All prototypes passed creative validation (VAD/DoP approval for aesthetic/storytelling alignment) and technical validation (TET confirmation of SRS/ICVFX compliance), with the window asset approved for the January 2026 test shoot.

V. CHALLENGES AND MITIGATION STRATEGIES

Cross-disciplinary collaboration between software engineering and filmmaking teams presented four core challenges, with actionable mitigation strategies implemented (Table 3). The primary lesson: proactive pre-project alignment (VP tool training, formal collaboration protocols, clear SRS) eliminates most gaps and accelerates delivery.

TABLE III

conclusion

需要我们补充 proof （证据） 关于本文提出了唯一的中国短片制作解决方案

后面的相关数据 也需要 proof

ref 补充证据相关的句子 really 的部分补充证据

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下面的自适应框架 requirements eng process 需要说明

VI. Conclusion

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This paper presents a context-aware adaptive software framework for AI-driven virtual production—the first tailored software engineering solution for smart filmmaking in Chinese short film production. Developed via cross-disciplinary collaboration between a software engineering team and a professional film crew, the framework embodies the core principles of smart system engineering (context perception, adaptive

proof?

decision-making, real-time execution) and integrates generative AI and ICVFX to address the unique challenges of smart filmmaking. The model-driven three-tier framework reduces AI-assisted 3D asset creation time by over 60%, maintains stable real-time ICVFX performance (≥24fps, 8,000–12,000 draw calls/frame), and effectively resolves cross-disciplinary collaboration gaps between software engineering and filmmaking teams.

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Empirical validation via a virtual quarantine room prototype confirms the framework's feasibility for small/medium Chinese film crews, enabling them to achieve international smart filmmaking production standards with limited resources. This work makes three key contributions to software engineering for smart systems and creative industries: (1) a scalable context-aware adaptive framework for AI-driven VP in smart filmmaking; (2) a formal requirement engineering process for translating unstructured creative requirements into verifiable smart system metrics; (3) actionable mitigation strategies for cross-disciplinary smart system development in creative industries.

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删掉 **acknowledgement** 和最后的结尾

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ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

RM

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